

3. Project Planning & Scheduling

Q.1 Explain the WBS

⇒

- The WBS is a method for getting a complex, multi-step project done in an easy and effective way by dividing it into smaller units so that you can get things done faster and more efficiently.
- Work Breakdown Structure (WBS) is a hierarchical tree structure that outlines your project and breaks it down into smaller and manageable units.
- Following are few reasons for creating WBS in a project:
 - 1) It creates an accurate and readable project organization
 - 2) It creates accurate assignment of responsibilities to the project team.
 - 3) It clearly lays down the project milestones and control points.
 - 4) It helps the project team to estimate the cost, time and risk.

- Characteristics

1) Definable

⇒ It is easily described and well understood by project participants.

2) Manageable

⇒ It is a meaningful project planning method where specific responsibility and authority are assigned to responsible individuals.

3) Independent

⇒ It allows minimum interface with or dependence on other ongoing elements.

4) Integration

⇒ It facilitates integration with other project work elements and with higher level cost estimates and schedule to include the entire project.

5) Measurable

⇒ WBS can be used to measure progress, has start and completion dates and measurable interim milestones.

• Different Forms of WBS

1) Phase-based structures

⇒ It defines and structures project activities based on project phases.

For e.g., Planning, Execution, closeout, etc.

2) Deliverable-based structures

⇒ It defines and structures project activities based on the deliverables agreed to deliver.

For e.g., user manual, unified CRM.

3) Responsibility-based structures

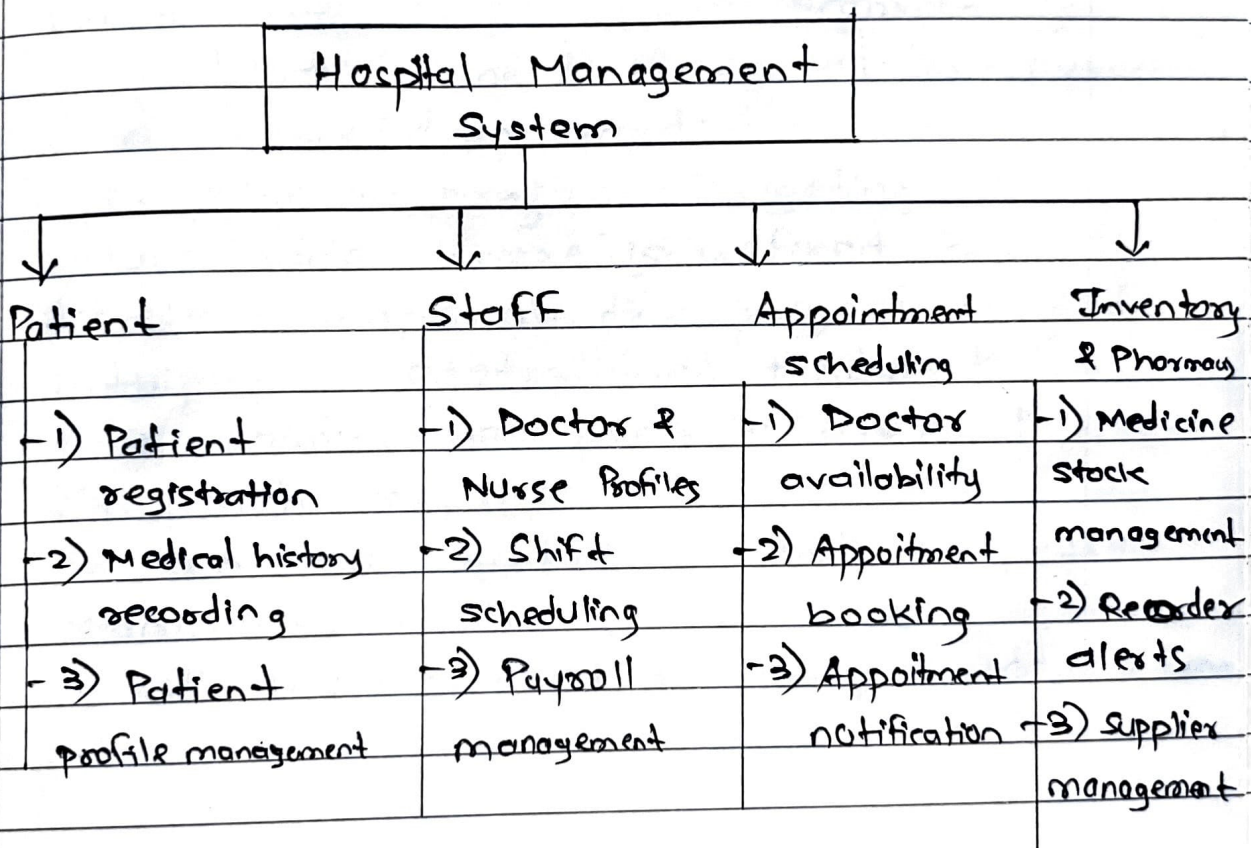
⇒ It defines and structures project activities based on the organization units that will work on the project.

- For eg., Project Management team, Core project team and testing team.

4) Resource Breakdown structure (RBS)

⇒ RBS is a PM tool that provides a hierarchical decomposition of resources.

• Example of Hospital Management System:



Structure of WBS

Level 1 : Project name (overall project)

Level 2 : Major deliverables or systems

Level 3 : Sub-components / tasks of each deliverables.

Level 4 : Further breakdowns until work is manageable.

Advantages of WBS

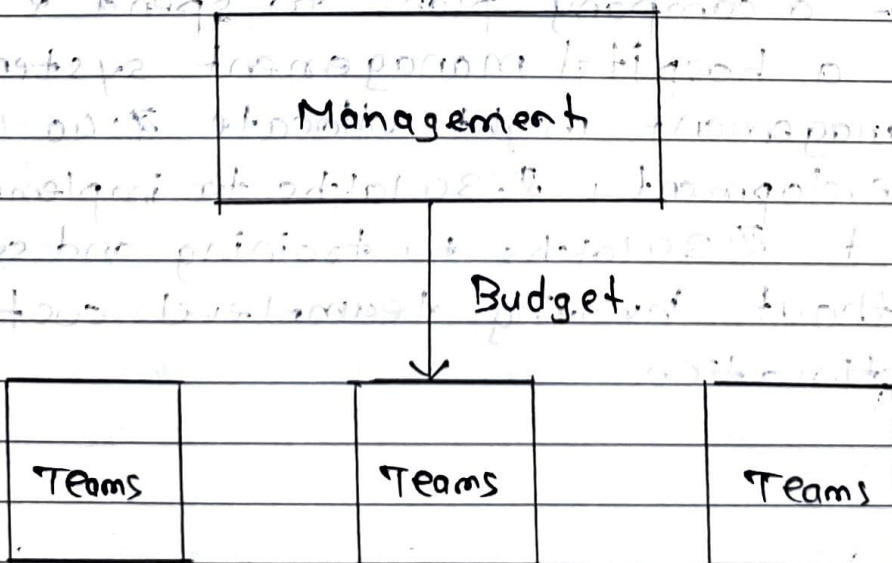
- 1) Improved planning
- 2) Smarter budget and resource allocation
- 3) Simplified risk identification
- 4) Enhanced visibility

Q.2 Explain top-down and bottom-up budgeting

⇒

1) Top-down Budgeting

- Top-down budgeting is a project estimation technique in which the overall project is estimated first, and in which the overall project is estimated first, and individual tasks are appointment from it.
- You start from the top of the pyramid and work downwards.
- This type of project budgeting usually occurs when there is a fixed budget.
- Also the scope of the project must fit within a predetermined funding level, particularly when projects aren't blessed with rich resources.
- In top-down budgeting, individual task estimates are only as accurate as the overall project estimate they are derived from.



Advantages

- 1) Faster budget creation
- 2) Strong alignment with overall company strategy
- 3) Easier to implement organization-wide changes
- 4) Provides clear direction from top management
- 5) Efficient for larger, more hierarchical organizations

Disadvantages

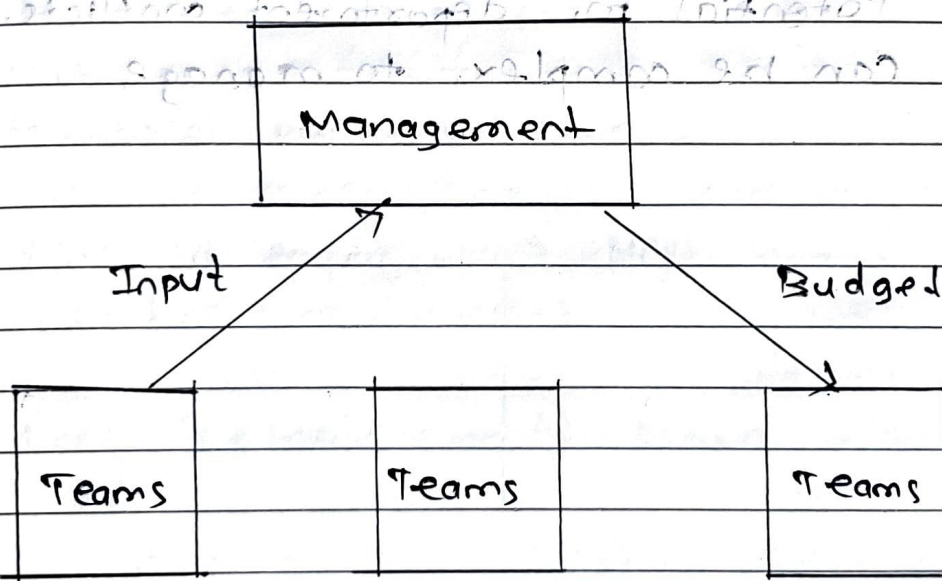
- 1) May overlook department-specific needs
- 2) Less detailed than bottom-up approach
- 3) Limited employee involvement and

For example

If a company plans to spend ₹ 1 crore on a hospital management system, management might allocate ₹ 40 lakhs to development, ₹ 30 lakhs to implementation and ₹ 30 lakhs to training and support without involving team-level cost estimation.

2) Bottom-up Budgeting

- ⇒ This method starts at the task level, where team members estimate the costs for each activity, which are then aggregated to form the total budget.
- Bottom-up estimating is the most accurate approach for estimating the cost and duration of project tasks.
 - This technique gives the entire project team the opportunity to take part in developing the estimates used to measure their work.



- So much information is available; considering all variables, the result can be extremely specific.
- Budgets can be changed according to current needs, and because of this, organizations can react quickly to change.
- For e.g., The development team estimates 12 lakh for frontend, 15 lakh for backend, 10 lakh for testing and so on.

Advantages :

- 1) Highly accurate and detailed budgets
- 2) Promotes employee engagement and ownership.
- 3) Better understanding of departmental needs.

Disadvantages :

- 1) Time-consuming process
- 2) Potential for department conflicts.
- 3) Can be complex to manage.

Q.3 Compare top-down budgeting and bottom-up budgeting.

⇒

Top-down Budgeting	Bottom-up Budgeting
1) Provides a broad overview, but lacks specific details about departmental needs.	1) Offers detailed insights from departments, resulting in higher accuracy.
2) Faster to implement as it focuses on high-level estimates.	2) More time-consuming.
3) Limited engagement.	3) High engagement.
4) Less flexible.	4) More adaptable.
5) May overlook specific resource needs of department.	5) Ensures resources are allocated based on detailed departmental requirement.
6) Suitable for larger organizations.	6) Best for smaller, agile projects/organization.
7) Can lead to miscommunication.	7) Promotes open communication.
8) Eg: Government infrastructure.	8) Eg: Software development.

Q.4 State various project estimation and scheduling techniques.

⇒

• Project Estimation Techniques

- 1) Top-down
- 2) Bottom-up
- 3) Analogous estimating
- 4) Three-point estimation (PERT)
- 5) Cost of quality.

1) Analogous Estimating

⇒ Here, experts who have experience in similar projects are approached for help.

- They use your own historical data.
- If you have relevant access to relevant historical data, try analogous estimating, which can show precedents that help define what your future costs will be in early stages of the project.

2) Three-point Estimate

⇒ Another approach is three-point estimate, which comes up with three scenarios: most likely, optimistic and pessimistic ranges.

- These are then put into an equation to develop cost estimation.

Formula:

$$\text{Expected time} = \frac{O + 4M + P}{6}$$

3) Cost OF Quality

- ⇒ This technique uses money spent during the project to avoid failures and money applied after the project to address failures.
- They can help fine-tune your overall project cost estimation.

• Project Scheduling Techniques

- 1) Gantt Chart
- 2) CPM
- 3) PERT
- 4) Agile Scheduling.

1) Gantt Chart

- ⇒ A gantt chart is a horizontal bar chart that visually represents a project plan over time.
- Modern Gantt charts typically shows you the status of as well as who's responsible for each task in the project.
- Ideal for monitoring process.

2) Critical Path Method

- ⇒ The CPM method helps you uncover the longest possible timeline for the project, as well as the shortest one.
- With the CPM you will also be able to mark critical tasks and the ones that may float.

- A critical path helps to visualize the project flow and calculate its duration when all dependencies and deliverables are known.

3) PERT

⇒ Similar to critical path, PERT is a way to visualize task's flow in the project and estimate the timeline based on their assumed duration.

- This technique also illustrates dependencies between tasks.

- In this approach, three time values are provide a measure of uncertainty associated with each activity -

i) Optimistic time (t_o)

ii) Most likely time (t_m)

iii) Pessimistic time (t_p)

$$t_c = t_o + 4 \cdot t_m + t_p$$

4) Agile scheduling

⇒ ~~Adjusts schedule based on resource~~

- Iterative and adaptive scheduling approach.

- Work is done in shorter sprints.

- Good for software & R&D projects.

Q.5 Short Note on: Gantt Chart.

⇒

- A Gantt bar chart is a horizontal bar chart that visually represents a project plan over time.
- Modern gantt charts typically shows you the status of as well as who's responsible for each task in the project.
- Key Parts of Gantt Chart:

1) Task list

⇒ Runs vertically down the left of the Gantt chart to describe project work and may be organized into groups and subgroups.

2) Timeline

⇒ Runs horizontally across the top of the Gantt chart and shows months, weeks, days and years.

3) Dateline

⇒ A vertical line that highlights the current date on the Gantt chart.

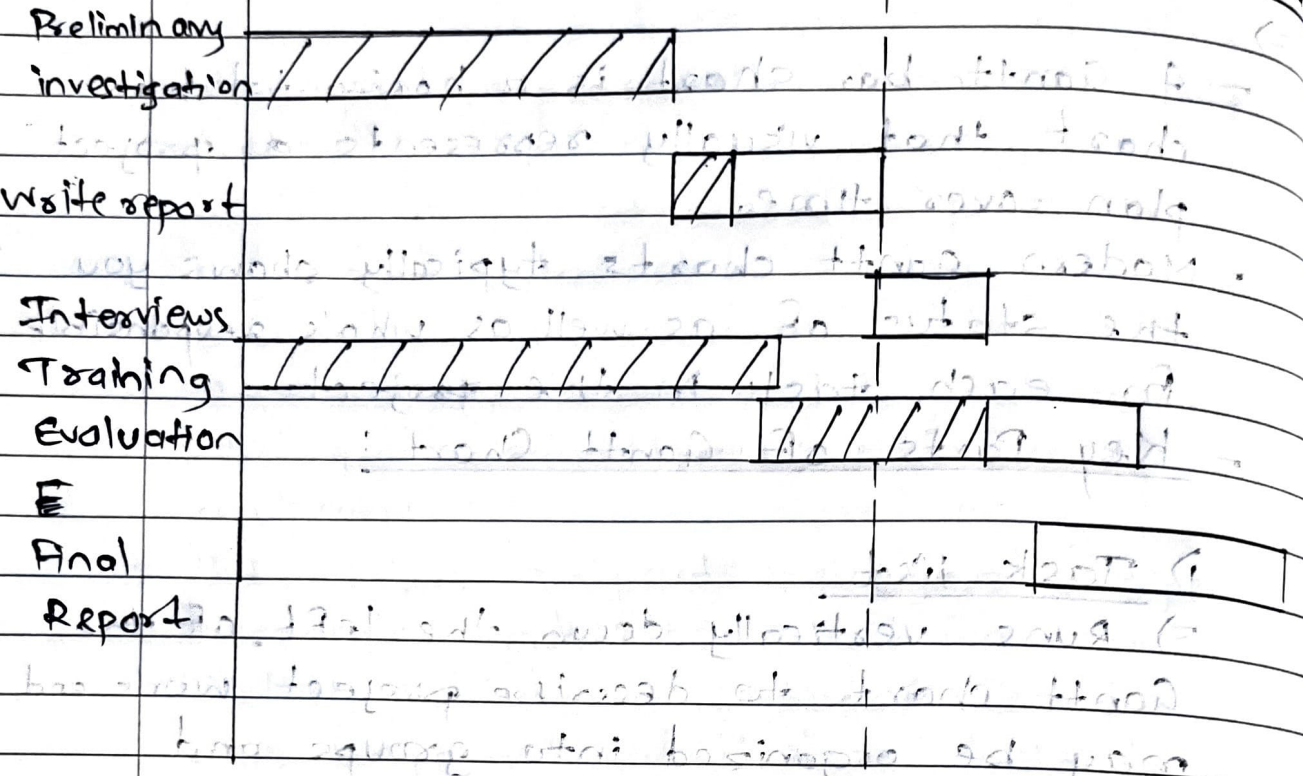
4) Milestone

⇒ Yellow diamonds that call out major events, dates, decisions and deliverables.

Report date



22/10 29/10 5/11 12/11 19/11 26/11 3/12 10/12 17/12 24/12



- Gantt chart comes in many forms, from good old-fashioned paper to desktop and even web-based software.
- Bringing these charts online transformed them from a static document that quickly becomes obsolete to a living, collaborative representation of project's current state.
- Gantt chart are useful in almost any industry.
- For eg.,
 - i) Construction
 - ii) Consulting agencies
 - iii) Marketing teams
 - iv) Manufacturing
 - v) HR

Benefits :

- 1) It provides a high-level overview.
- 2) It improves efficiency and helps manage resources.
- 3) It allows for better tracking.
- 4) It illustrates overlaps and dependencies.
- 5) It boosts productivity.

Limitations :

- 1) They can become extraordinarily complex.
- 2) They need to be constantly updated.
- 3) Setup can be time-consuming.
- 4) They don't designate priorities.

Q.6 Note on : PMIS.

⇒

- A project management information system (PMIS) - is a computer-based information system that efficiently stores and organized information needed to run a project.
- It collects and uses project information through one or more software applications.
- It helps project managers to plan, execute and close their project.
- It is used to support all aspects of the project from initiating through closing and can include both manual and automated systems.
- PMIS is made to support all aspects of project management by collecting and storing information.
- Some of those areas are integration management, project scope management, project cost management, project time management, etc.
- In a project there is a lot of information to keep track of and while managing a project, it is crucial to receive information required at that moment.
- Therefore, a PMIS is becomes important.
- It's the tool that gives you instant access to the required information.

- This information is also critical for future projects in terms of reducing risks, improving efficiencies and lowering costs.

- A PMIS captures all project data and stores them in an organized way. It also makes them retrievable, searchable, categorized and analytics.

• Roles :

1) During the initiation phase, a project manager uses PMIS to help prepares a preliminary budget including cost estimates and resources.

2) PMIS in conjunction with other tools helps with scheduling the project.

3) When planning, PMIS helps with detailing the scheduling including task and critical path.

4) It supports cost management planning including WBS analysis and integration of control processes.

5) Enables seamless communication among teams across locations.

6) Reduces manual errors through automation.

- PMIS uses a series of tools:

1) Scheduling and Planning

⇒ Computes early and late schedule, slack times and the critical path.

2) Resource management

⇒ It including resource loading, levelling, allocation, etc.

3) Budget

⇒ It associates cost with individual tasks for more accurate budget estimation and generation.

4) Control and Performance

⇒ It analyzes and controls cost and performance, update existing plans as actual against planned data changes and provides what-if scenarios for the project manager.

5) Reporting & Communication

⇒ It facilitates creation of graphs and charts of collected and analyzed data that can be shared with stakeholders and team members.

Q.7 (Dec 24) (Q4.a)

⇒

Activity	Predecessor Activity	t_o	t_m	t_p
A	-	2	4	6
B	A	3	6	9
C	A	8	10	12
D	B	9	12	15
E	C	8	9	10
F	D, E	16	21	26
G	D, E	19	22	25
H	F	2	5	8
I	G	1	3	5

1) Draw the project network

⇒

Activity	Predecessor Activity	t_o	t_m	t_p	t_e
A	-	2	4	6	4
B	A	3	6	9	6
C	A	8	10	12	10
D	B	9	12	15	12
E	C	8	9	10	9
F	D, E	16	21	26	21
G	D, E	19	22	25	22
H	F	2	5	8	5
I	G	1	3	5	3

$$\therefore t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$t_A = \frac{2 + 4(4) + 6}{6} = 4 \text{ days}$$

$$t_B = \frac{3 + 4(6) + 9}{6} = 6 \text{ days}$$

$$t_C = \frac{8 + 4(10) + 12}{6} = 10 \text{ days}$$

$$t_D = \frac{9 + 4(12) + 15}{6} = 12 \text{ days}$$

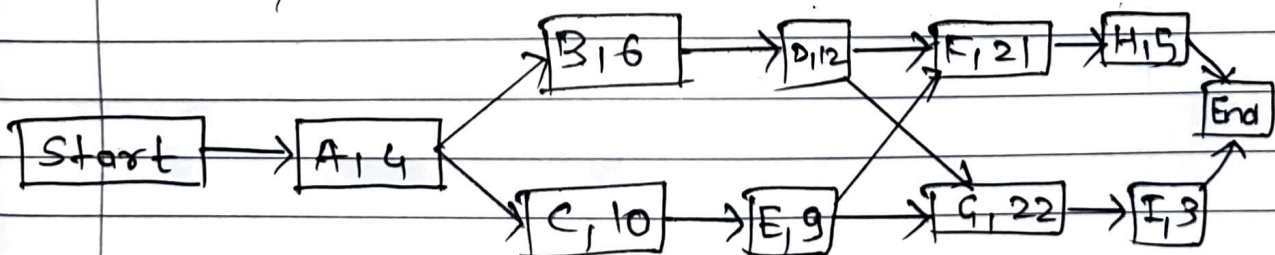
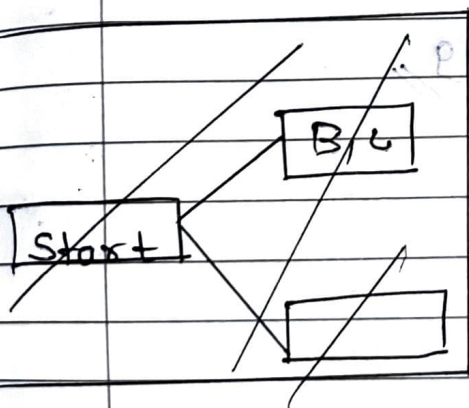
$$t_E = \frac{7 + 4(8) + 9}{6} = 9 \text{ days}$$

$$t_F = \frac{16 + 4(21) + 26}{6} = 21 \text{ days}$$

$$t_G = \frac{21 + 4(22) + 25}{6} = 22 \text{ days}$$

$$t_H = \frac{2 + 4(5) + 8}{6} = 5 \text{ days}$$

$$t_I = \frac{7 + 4(3) + 5}{6} = 3 \text{ days}$$



2) Critical Path

i) Start $\rightarrow$ A $\rightarrow$ B $\rightarrow$ D $\rightarrow$ F $\rightarrow$ H $\rightarrow$ End
 $\Rightarrow te = 4 + 6 + 12 + 21 + 5 = 48$

ii) Start $\rightarrow$ A $\rightarrow$ B $\rightarrow$ D $\rightarrow$ G $\rightarrow$ I $\rightarrow$ End
 $\Rightarrow te = 4 + 6 + 12 + 22 + 3 = 47$

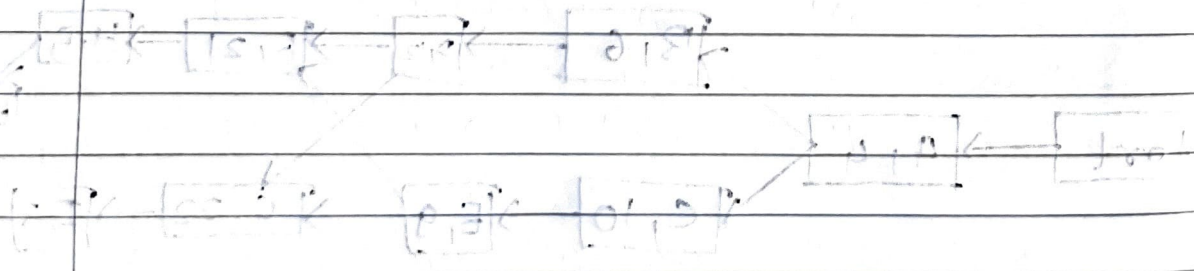
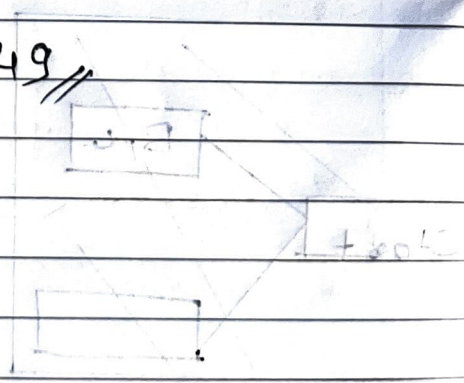
iii) Start $\rightarrow$ A $\rightarrow$ C $\rightarrow$ E $\rightarrow$ G $\rightarrow$ I $\rightarrow$ End
 $\Rightarrow te = 4 + 10 + 9 + 22 + 3 = 48$

iv) Start $\rightarrow$ A $\rightarrow$ C $\rightarrow$ E $\rightarrow$ F $\rightarrow$ H $\rightarrow$ End
 $\Rightarrow te = 4 + 10 + 9 + 21 + 5 = 49$ ✓

Now, maximum time = critical path.

Start $\rightarrow$ A $\rightarrow$ C $\rightarrow$ E $\rightarrow$ F $\rightarrow$ H $\rightarrow$ End

3) time required = 49



Q.8 (May 24) Q2.6

⇒ Solⁿ:-

Activity	Predecessor	Duration			t_e	σ	Var	slack
		t_o	t_m	t_p				
A	—	2	4	12	5	1.67	2.7889	0
B	—	10	12	26	14	2.67	7.1289	1
C	A	8	9	10	9	0.33	0.1089	1
D	A	10	15	20	15	1.67	2.7889	0
E	A	7	7.5	11	8	0.67	0.4489	1
F	B, C	9	9	9	9	0	0	1
G	D	3	3.5	7	4	0.67	0.4489	0
H	E, F, G	5	5	5	5	0	0	0

Z	Probability
0	0.5
0.5	0.6950
1	0.843

$$\therefore t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$\therefore t_A = \frac{2 + 4(4) + 12}{6} = 5$$

$$\therefore t_B = \frac{10 + 4(12) + 26}{6} = 14$$

$$\therefore t_C = \frac{8 + 4(9) + 10}{6} = 9$$

$$\therefore t_D = \frac{10 + 4(15) + 20}{6} = 15$$

Similarly for E, F, G, H

$$\therefore \sigma = \frac{12 - 2}{6} = 1.67$$

$$\therefore \sigma_A = \frac{12 - 2}{6} = 1.67$$

$$\therefore \sigma_B = \frac{26 - 10}{6} = 2.67$$

Similarly for others.

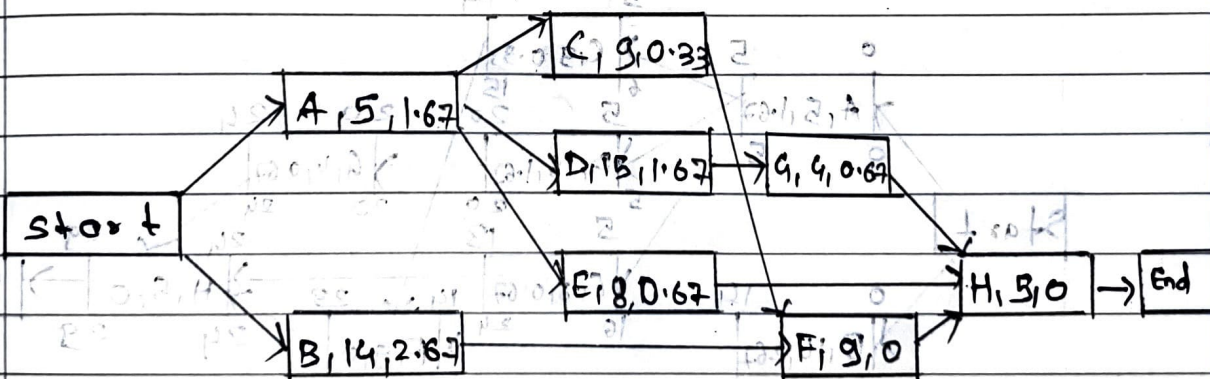
Now,

$$\sigma_A^2 = (1.67)^2 = 2.7889$$

$$\sigma_B^2 = (2.67)^2 = 7.1289$$

and so on.

1) Network diagram :



2) Critical Path and expected project duration :

- Total paths :

1) Start $\rightarrow$ A $\rightarrow$ C $\rightarrow$ F $\rightarrow$ H $\rightarrow$ End

$$t_e = 5 + 9 + 9 + 5 = 28 \text{ weeks}$$

ii) Start $\rightarrow$ A $\rightarrow$ D $\rightarrow$ G $\rightarrow$ H $\rightarrow$ End

$$t_e = 5 + 13 + 4 + 5 = 27 \text{ weeks.}$$

iii) Start $\rightarrow$ A $\rightarrow$ E $\rightarrow$ H $\rightarrow$ End

$$t_e = 5 + 8 + 5 = 18 \text{ weeks.}$$

iv) Start $\rightarrow$ B $\rightarrow$ F $\rightarrow$ H $\rightarrow$ End

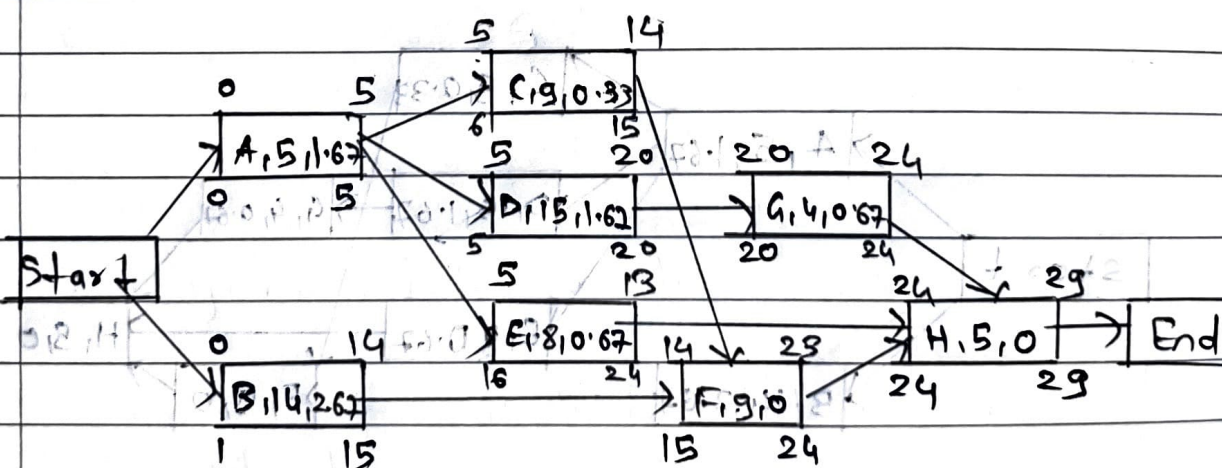
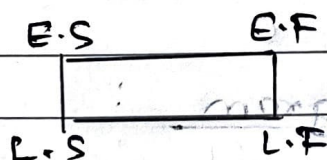
$$t_e = 14 + 9 + 5 = 28 \text{ weeks}$$

Now, maximum the week/time = Critical Path

$\therefore$ Start $\rightarrow$ A $\rightarrow$ D $\rightarrow$ G $\rightarrow$ H $\rightarrow$ End $\rightarrow$ Critical

Path

• Slovak



$$\text{Slovak} = E.S - L.S \quad \text{OR} \quad E.F - L.F$$

iv) Probability :

$$Z = \frac{\sigma_1 - \mu}{\sigma_{cp}}$$

σ_1 = duration given
 μ = actual duration
 $\sigma_{cp} = \sqrt{\text{critical path mean}}$

$$\sigma_{cp} = \sqrt{\sigma_A^2 + \sigma_D^2 + \sigma_G^2 + \sigma_H^2}$$

$$= 812.48549$$

$$\therefore Z = \frac{30 - 29}{\sqrt{812.48549}}$$

$$= 0.4073$$

$\therefore$ Probability of 30 weeks

$$= P(0.4073) = \text{SDV} \therefore$$

$$= 0.6581 \times 100$$

$$= 65.81\%$$

Q.9 (May 23) Q.4 a

⇒

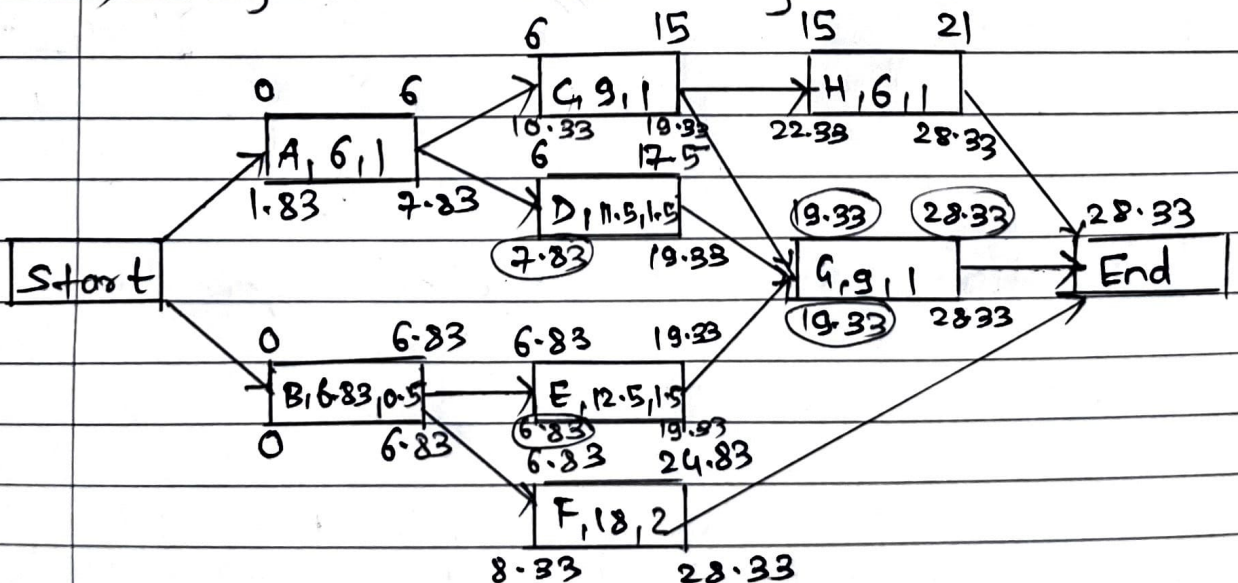
Activity	Predecessor	to	tm	tp	te	σ	vor	slack
A	-	3	6	9	6	1	1	1.83
B	-	5	7	8	6.83	0.5	0.25	0
C	A	6	9	12	9	1	1	4.83
D	A	6	12	15	11.5	1.5	2.25	1.83
E	B	9	12	18	12.5	1.5	2.25	0
F	B	12	18	24	18	2	4	3.50
G	C, D, E	6	9	12	9	1	1	0
H	C	3	6	9	6	1	1	7.33

$$\therefore te = \frac{to + 4tm + tp}{6}$$

$$\therefore \sigma = \frac{tp - to}{6}$$

$$\therefore vor = \sigma^2$$

1) Project Network diagram:



Paths :

i) Start $\rightarrow$ A $\rightarrow$ C $\rightarrow$ H $\rightarrow$ End

$$t_e = 6 + 9 + 6 = 21$$

ii) Start $\rightarrow$ A $\rightarrow$ D $\rightarrow$ G $\rightarrow$ End

$$t_e = 6 + 11.5 + 9 = 26.5$$

iii) Start $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G $\rightarrow$ End

$$t_e = 6 + 9 + 9 = 24$$

iv) Start $\rightarrow$ B $\rightarrow$ E $\rightarrow$ G $\rightarrow$ End

$$t_e = 6.83 + 12.5 + 9 = 28.33$$

v) Start $\rightarrow$ B $\rightarrow$ F $\rightarrow$ End

$$t_e = 6.83 + 18 = 24.83$$

Start $\rightarrow$ B $\rightarrow$ E $\rightarrow$ G $\rightarrow$ End $\rightarrow$ critical path

3) Find probability between 26 to 31 days.
 $\Rightarrow$

$$\begin{aligned} Z_{26} &= \frac{21 - \mu}{\sqrt{\sigma_{cp}}} \\ &= \frac{26 - 28.33}{\sqrt{\sigma_B^2 + \sigma_E^2 + \sigma_G^2}} \\ &= \frac{2.83}{1.87} \\ &= -1.2460 \end{aligned}$$

$$\therefore P(26) = 0.1064 \times 100 = 10.64\%$$

$$Z_{31} = \frac{31 - 28.33}{1.87}$$

$$= 1.4278$$

$$\therefore P(31) = 10.9233\% \times 100 = 10.92\%$$

$$\therefore P(31) = 10.9233\%$$

$$P(31) = P + 2.1149 = 10.92\%$$

$$\therefore P(26 \leq P \leq 31) = 92.33\% - 10.64\%$$

$$PS = P + P = 81.69\%$$

$$b_{n3} \leftarrow 0 \leftarrow 2 \leftarrow 8 \leftarrow 1.0062$$

$$= 0.89 + 2.51 + 5.8.8 = 9.1$$

$$2) \underline{\lambda = 30}$$

$$b_{n3} \leftarrow 1 \leftarrow 8 \leftarrow 1.0062$$

$$E_{22} = 30.8 + 28.33 = 9.1$$

$$1.87$$

$$= 0.8930$$

$$b_{n3} \leftarrow 0 \leftarrow 3 \leftarrow 8 \leftarrow 1.0062$$

$$\therefore P = P(0.8930) \times 100$$

$$= 0.8141 \times 100$$

$$= 81.41\%$$

$$N = 10$$

$$1.87$$

$$88.85 - 28.33 =$$

$$60.52$$

$$88.85$$

$$1.87$$

$$0.8141 =$$

$$0.01 \times 0.001 = 0.00001$$

1 Explain Linear Responsibility Chart (LRC) with an example.

- LRC is the chart of responsibility which identifies the project participants and shows authority and responsibility relationship among them.

- The participants may be general manager, managers of projects, project manager and functional manager.

- It clearly specifies the authority and responsibility relationships of project participants to avoid confusion and conflicts.

- The LRC promotes communication and coordination of responsibilities within a project.

- The LRC is divided into:

- 1) Rows - They indicate activities, authority, responsibilities.
- 2) Columns - They identify position of project participants.
- 3) Numbers - They indicate the degree of authority-responsibility existing between the rows and columns.

- LRC is also known as Linear Chart (LC), Matrix Responsibility Chart (MRC), Responsibility Interface Matrix (RIM), Responsibility and Accountability (RAM).

For e.g.,

Let's consider a software development project with key tasks and team roles.

Task	PM	Developer	Tester	Client
Requirement Analysis	A	R	-	C
Design UI/UX	C	R	-	I
Develop applications	C	R	-	I
Testing & Bug Fixing	I	R	A	-
Final Approval	A	-	-	R

- Key responsibilities codes:

R (Responsible): The person who executes the task.

A (Accountable): The person ultimately answerable for task completion.

C (Consulted): The person providing input or expertise.

I (Informed): The person who is kept updated on progress.

- Advantages:

- 1) Avoids confusion about who is responsible for what.
- 2) Better collaboration.
- 3) Communication is facilitated.

- Disadvantages:

1) All relationships may be difficult to delineate.

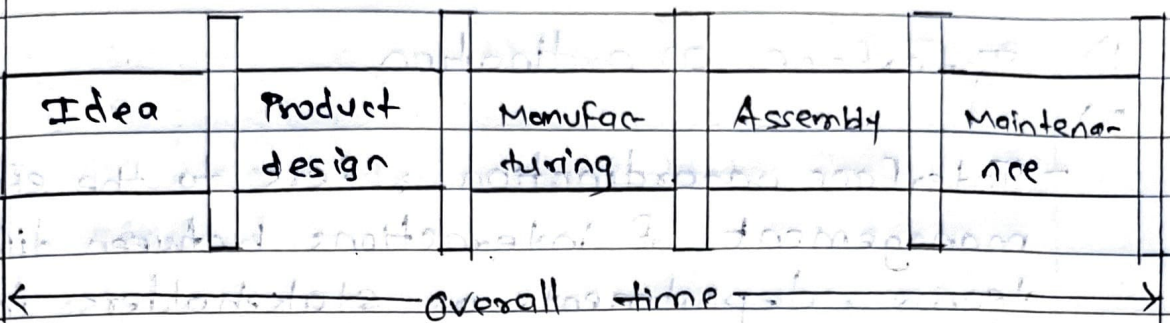
2) It does not describe the people's interaction in the project.

Write short notes

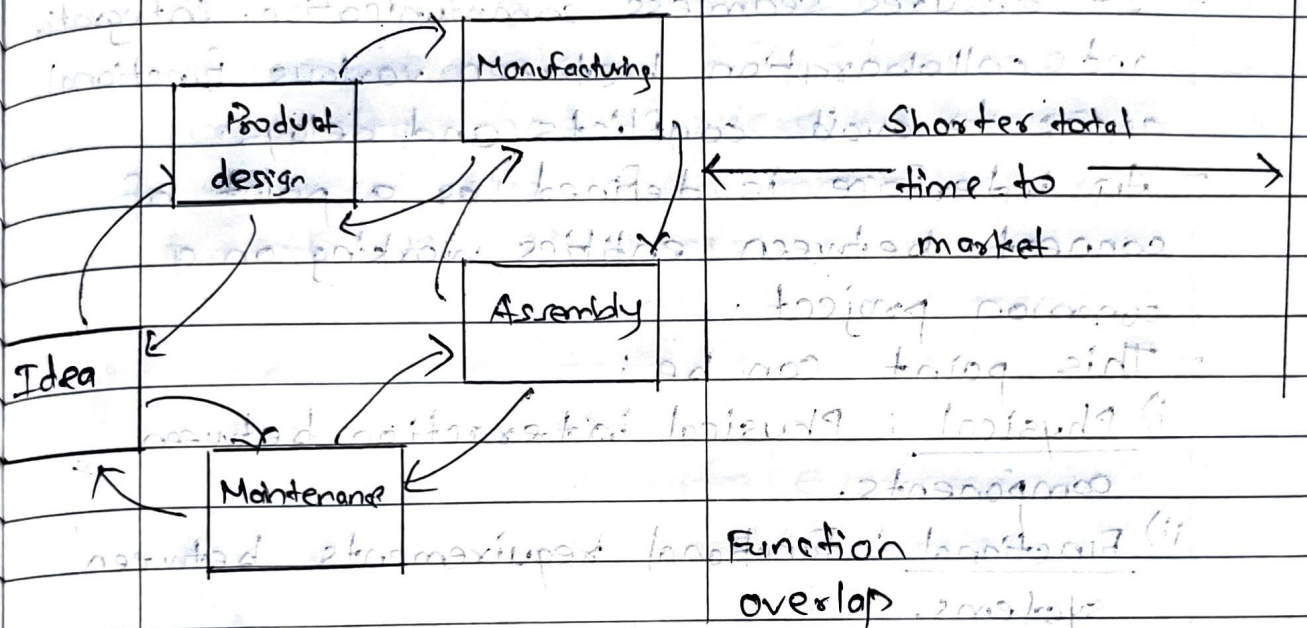
Concurrent Engineering

- Concurrent Engineering (CE) has emerged as a discipline to help achieve the objectives of reduced cost, better quality and improved delivery performance.
- Concurrent engineering is a systematic approach to integrated product development that emphasises the response to customer expectations.
- Key Features :-
 - 1) Parallel execution of tasks instead of step-by-step approach.
 - 2) Collaboration between cross-functional teams.
 - 3) Early identification and resolution of design flows.
 - 4) Faster product development cycles.
- Following figure compares the traditional Sequential Approach with the Concurrent Engineering approach in product development.
- In the traditional approach, each phase occurs one after another (Idea → product design → Manufacturing → Assembly → Maintenance).
- The overall time to complete the project is longer.
- There is a less collaboration between different teams, leading to inefficiencies.

Partial Processes



Concurrent Product Design Approach



- Now in concurrent engineering phase, instead of waiting for one phase to finish before starting the next, multiple phases occurs simultaneously.
- In this, teams works together, reducing errors and improving quality.
- Since processes overlap, the product reaches the market much faster.

b) Interface co-ordination

⇒

- Interface co-ordination refers to the effective management of interactions between different teams, departments or stakeholders in a project.
- It ensures seamless communication, integration and collaboration between various functional areas to avoid conflicts and delays.
- An interface is defined as a point of connect between entities working on a common project.
- This point can be:
 - i) Physical: Physical interaction between components.
 - ii) Functional: Functional requirements between systems.
 - iii) Contractual: Interaction between subcontractors or suppliers.
 - iv) Organizational: Information exchanged between disciplines.
 - v) Knowledge: General information exchanged between parties.
 - vi) Resource: Points of dependencies between equipment, material and labour suppliers.
- Roles and Responsibilities of Interface Co-ordinator:
 - i) Participate in interface meetings with involved parties as required to manage interfaces.

2) Ensure resolution of inbound interface requests.

3) Provide collaborative interface resolution support.

- For e.g.,

In construction projects, interface coordination is essential between architects, engineers, contractors and suppliers to ensure proper alignment of design, materials and construction timelines.

• Types of interface coordination:

1) Internal Interface

→ Where the complete responsibility lies within Project team scope of work.

2) External Interface

→ These ~~part~~ are the part of scope of work under the responsibility of contractor related to the project.